

Technical Security Document: Tarpit Trap Architecture

Project: ASHA Saathi AI **Module:** Backend Active Cybersecurity **Component:** Automated Bot & Malicious Code Defense **Date:** 17 July 2026

1. Executive Summary

Ek 'Tarpit' trap cybersecurity ka ek *advanced defense mechanism* hai. Aam firewalls attackers ko seedha block karke unhe alert kar dete hain, jabki ek tarpit unhe block karne ke bajay backend system mein fasa leta hai. ASHA Saathi AI mein jahan patients ka *sensitive medical data aur biometric security* shamil hai, yeh active defense layer malicious code aur data-scraping bots ko database tak pahunchne se pehle hi rokne mein madadgar sabit hogi.

2. Implementation Architecture (4-Step Process)

Step 1: The Bait (Jaal Bichhana)

Backend system mein aisi *fake API routes (endpoints)* banayi jayengi jo ASHA workers ya doctors ke kisi kaam ki nahi hoti, lekin attackers aur bots inki talash mein rehte hain.

- **Examples:** `/api/admin_panel`, `/wp-login.php`, ya `/api/v1/database/dump`.
- **Logic:** Ek normal user kabhi in links par click nahi karega. Agar koi in endpoints par request bhej raha hai, toh yeh *100% ek automated bot, scraper, ya malicious script* hai.

Step 2: The Trap (Fasana)

Aam web servers aisi galat request aane par turant *404 Not Found* ya *403 Forbidden error* de dete hain, jisse attacker ko pata chal jata hai ki route exist nahi karta.

- **Tarpit Logic:** Node.js (Express) mein ek custom middleware likha jayega jo error dene ke bajay us bot ka HTTP connection apne paas **'open' (hold)** kar lega.

Step 3: The Tar (Slow-down Effect)

Attacker ke system ko *slow aur exhaust* karne ke liye connection ko jaan-boojh kar atkaya jayega.

- **Execution:** Hamara backend server har 10-15 second mein attacker ko sirf *1 byte data* (jaise ek khali space " ") bhejega.
- **Impact:** Kyunki connection open hai aur data dheere-dheere aa raha hai, hacker ka script us connection ko close nahi kar payega. Agar wo 100 requests bhejega, toh uske *100 server threads block* ho jayenge. Attacker ka CPU aur RAM is fake request ka wait karne mein barbad hone lagega.

Step 4: The Catch (Block & Report)

Jab tak attacker is digital jaal (tarpit) mein fasa hua struggle kar raha hoga, usi waqt background mein humara security system apna final kaam karega:

- **Logging:** Us malicious bot ka IP address, browser detail (User-Agent), aur location turant log kar li jayegi.
- **Blacklisting:** Is IP address ko automatically Supabase ke RLS (Row Level Security) rules aur Cloud Firewall mein "**Blacklist**" kar diya jayega.
- **Result:** Jab tak bot connection band karke naya rasta dhundhne ki koshish karega, tab tak use ASHA Saathi ke asli system aur servers se hamesha ke liye block kiya ja chuka hoga.